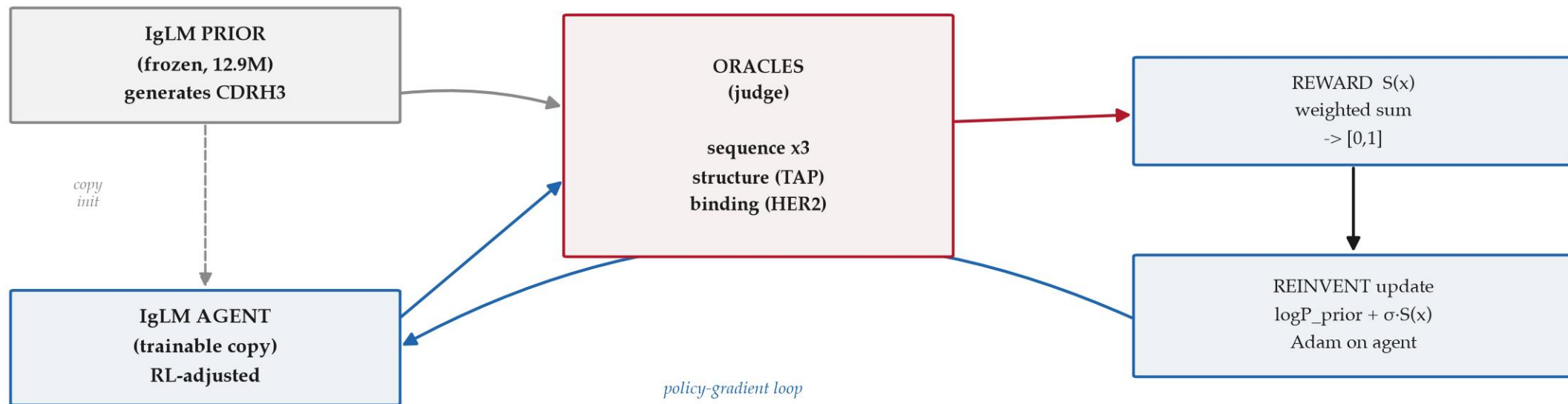
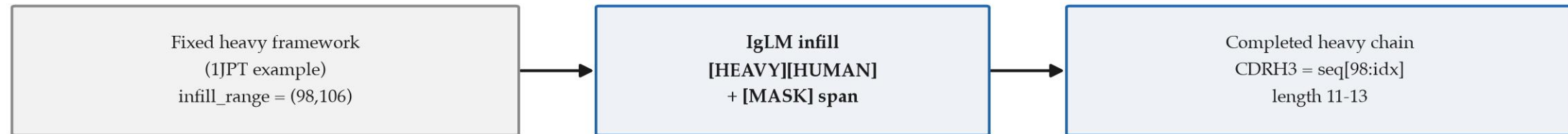


Overall pipeline: prior-guided RL with a multimodal judge



Step 1 — Generator: IgLM infill of CDRH3 on a fixed framework



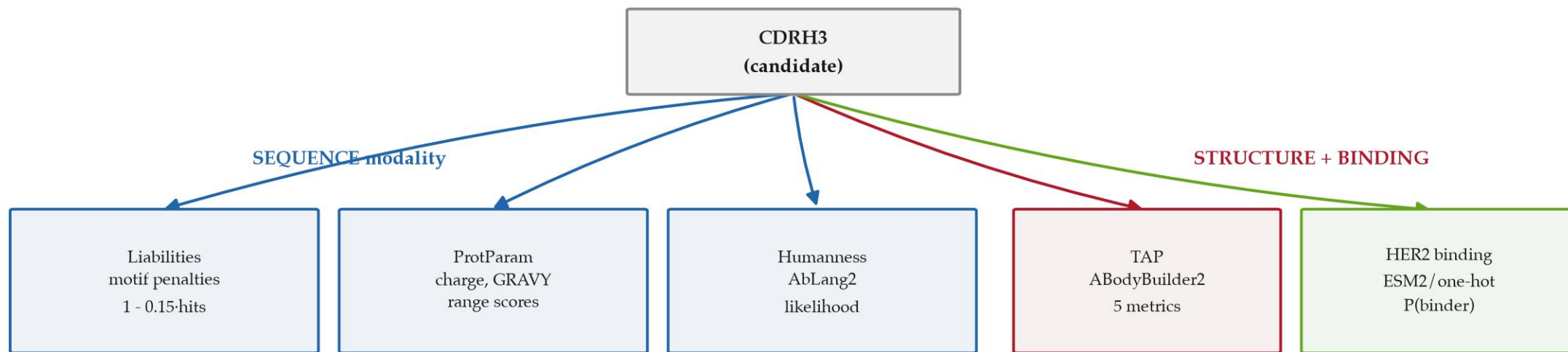
Infill token layout (recreated for the differentiable likelihood):

$[\text{HEAVY}][\text{HUMAN}] \text{fw}_{\text{left}} [\text{MASK}] \text{fw}_{\text{right}} [\text{SEP}] x_1 x_2 \cdots x_L [\text{CLS}]$

$\text{CDRH3 span} = x_1 \dots x_L$ (generated after [SEP])

Scoring reads that span plus the closing [CLS] token.

Step 2 — Oracles: a multimodal judge, each maps a sequence to [0,1]



Per-oracle scores feed the weighted property score:

$$S(x) = 0.50 \text{ human}(x) + 0.30 \text{ liab}(x) + 0.20 \text{ physchem}(x) \in [0, 1]$$

$$\text{TAP score} = \frac{1}{5} \sum_m \text{flag}(m), \text{ flag} \in \{0, 0.5, 1\} \text{ (green / amber / red)}$$

Oracles are near-orthogonal: pairwise $|r| < 0.06$ on the reference pool.

Step 3 — Reward and the REINVENT update



Augmented log-likelihood (the RL target):

$$\log P^*(x) = \log P_{\text{prior}}(x) + \sigma S(x)$$

Loss regressed on the agent (the prior term is the leash):

$$\mathcal{L}(x) = (\log P_{\text{agent}}(x) - [\log P_{\text{prior}}(x) + \sigma S(x)])^2$$

σ trades property gain against staying antibody-like. Guard: CDRH3 < 4 residues $\Rightarrow S = 0$.

Step 4 — RL loop: the differentiable CDRH3 log-likelihood



Span log-likelihood, kept differentiable for the agent:

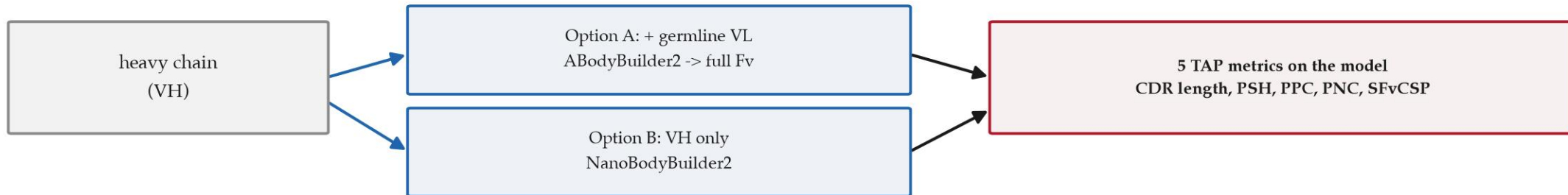
$$\log P(\text{CDRH3} \mid \text{framework}) = \sum_{t \in \text{span}} \log p_{\theta}(x_t \mid x_{<t})$$

agent: requires_grad
(gradient flows)

prior: no_grad
(frozen target)

Validated against IgLM native log_likelihood: mean -2.0315 == -2.0315 (match to 1e-3).

Step 5 — TAP structural oracle (the multimodal step)



Surface metrics weight each CDR-vicinity residue by its solvent accessibility (SASA):

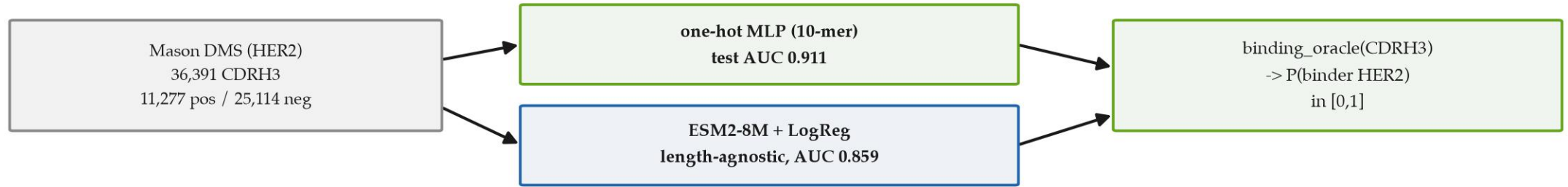
$$\text{PSH} = \sum_{i \in \text{CDR vic}} \text{KD}(a_i) \cdot \text{SASA}_i / 100$$

$$\text{SFvCSP} = q_{\text{net}}(V_H) \times q_{\text{net}}(V_L) \quad (\text{needs both chains} \rightarrow \text{Option A only})$$

Used as a final filter on the RL shortlist. Sequence and structure are orthogonal (r about 0),

so TAP reorders candidates the sequence oracles rank equally. TAP-style: reimplemented from the paper.

Step 6 — HER2 binding oracle and its explainability



XAI: in-silico saturation mutagenesis around wild-type trastuzumab CDRH3

$$\Delta_{p,a} = P(\text{binder} \mid \text{WT with } x_p = a) - P(\text{binder} \mid \text{WT})$$

sensitivity(p) = $\text{std}_a P(\text{binder} \mid x_p = a)$ over the 20 substitutions

Result: WT scores 0.97 and sits at the optimum at every sensitive position;

position 8 is the binding hotspot. Confirms the oracle learned real signal.

Honest scope: transferred proxy (Mason length-10 trastuzumab vs 11-13mer on another framework).